

40.6 Summary and Notes

40.6.1 Summary

Idea.

1. A nonlinear link and a random effect do not commute, so a clustered model has two sets of coefficients. The identity and log links are the exceptions; for the probit the marginal coefficient is $\beta^C / \sqrt{1 + \tau^2}$ exactly, and for the logit it is strictly attenuated with local factor $1 - \text{Var}(P_\theta) / \{E P_\theta E(1 - P_\theta)\}$ (Proposition 40.1.1).
2. The generalized linear mixed model puts random effects in the linear predictor of a generalized linear model (Definition 40.2.1). Its marginal moments follow by conditioning: a binomial response acquires the beta-binomial overdispersion factor, and a Poisson–lognormal response acquires *exactly* the NB2 variance function (Proposition 40.2.1).
3. A cluster-level random intercept can only induce nonnegative within-cluster correlation, because two nondecreasing functions of one random variable have nonnegative covariance (Lemma 40.2.2). Data with competition inside a cluster need a different tool.
4. Predicting a random effect in a GLMM is empirical Bayes rather than BLUP, but the shrinkage survives: the posterior mode lies strictly between zero and the cluster’s own estimate, with the familiar factor $\tau^2 / (\tau^2 + I_i^{-1})$ to first order (Proposition 40.2.3).
5. The integrated likelihood factorizes over independent clusters. The Laplace approximation has relative error $O(n_i^{-1})$ — governed by the observations *per cluster*, not by the sample size — and adaptive Gauss–Hermite quadrature with $K = 1$ node is exactly the Laplace approximation (Theorem 40.3.1).
6. Penalized quasi-likelihood is Henderson’s mixed-model equations applied to the working response, and is inconsistent: it attenuates both $\hat{\beta}$ and $\hat{\theta}$, badly for binary data in small clusters (Proposition 40.3.2, Example (Three methods on clustered binary data)).
7. Generalized estimating equations solve $\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ with a working correlation the analyst chooses and does not have to believe (Definition 40.4.1). The estimating function is unbiased for every working correlation, so $\hat{\beta}$ is consistent for the marginal coefficient and asymptotically normal with the sandwich covariance $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}$ (Theorem 40.4.1).
8. GEE gives no likelihood, no random-effect predictions and no distribution; a non-diagonal working correlation also needs the mean model to hold conditionally on the *whole* covariate history, which feedback destroys; and it fails under missingness that depends on previously observed responses — the one case in which a likelihood-based analysis still works (Section 40.4).
9. A wrong working correlation costs efficiency for covariates that vary within clusters and nothing at all for covariates fixed within a cluster; the sandwich needs a small-sample correction below about forty clusters; and QIC, the quasi-likelihood analogue of AIC, is sound for mean models and weak for correlation structures (Proposition 40.5.1).
10. Because every working correlation targets the same β^0 , a large gap between two

GEE fits of the same mean model is evidence against the *mean model* (Example (Grunfeld’s panel by GEE)).

40.6.2 Notes and sources

Coverage. The topics follow Agresti (2015, sections 9.4–9.6) and Fahrmeir, Kneib, Lang and Marx (2021, sections 7.5–7.6), but put the marginal–conditional distinction first, before either estimation method, because that is where the choice between the two traditions is actually made. Book-length treatments are Diggle, Heagerty, Liang and Zeger (2002) for longitudinal data, McCulloch, Searle and Neuhaus (2008) for the mixed-model theory, and Molenberghs and Verbeke (2005) for discrete longitudinal responses.

Marginal versus conditional. The distinction was made precise for binary data by Neuhaus, Kalbfleisch and Hauck (1991), still the clearest statement of what each coefficient means and when the two agree. The approximation factor $(1 + c^2\tau^2)^{-1/2}$ with $c = 16\sqrt{3}/(15\pi)$ is from Zeger, Liang and Albert (1988); the constant comes from matching the logistic and normal distributions and predates its use here by a long way. The proof of Proposition 40.1.1(d) given above, through the derivative identity Eq. 40.1.5, seems to be the shortest route to the strict inequality, and needs no assumption on the random-effect distribution beyond non-degeneracy. The median odds ratio of Exercise (B1) was popularized in epidemiology by Larsen and Merlo (2005).

Fitting. Breslow and Clayton (1993) introduced penalized quasi-likelihood under that name and set the pattern for GLMM software for a decade; Breslow and Lin (1995) and Lin and Breslow (1996) quantified its bias and proposed corrections. Adaptive Gauss–Hermite quadrature for this problem is due to Liu and Pierce (1994) and, for nonlinear mixed models, Pinheiro and Bates (1995), who also observed that one adaptive node reproduces the Laplace approximation. The Laplace expansion itself is classical; the form used in Theorem 40.3.1(b) is the one in Tierney and Kadane (1986) and Barndorff-Nielsen and Cox (1989, chapter 3). The asymptotics of the integrated likelihood quoted in Section 40.2 are in McCulloch, Searle and Neuhaus (2008, chapter 14). Monte Carlo EM for this problem is Wei and Tanner (1990), with automated sample-size rules in Booth and Hobert (1999); the MCMC route with iteratively reweighted proposals is Gamerman (1997).

Estimating equations. Liang and Zeger (1986) and Zeger and Liang (1986) introduced GEE, and the pair is worth reading together: the first gives the theory, the second the practice. The conditions and the expansion in Theorem 40.4.1 follow theirs, with the cluster sandwich argument taken from Theorem 21.4.2 as Section 33.2 already arranged it. The warning about time-varying covariates and the full-covariate conditional mean is due to Pepe and Anderson (1994), and is treated at length in Diggle, Heagerty, Liang and Zeger (2002, chapter 12). Small-sample corrections to the sandwich are due to Mancl and DeRouen (2001), Kauermann and Carroll (2001) and Fay and Graubard (2001); the first is used in Example (Coverage when the working correlation is wrong). QIC is Pan (2001); the criticism of it as a selector of working correlations, and the CIC variant, are in Hin and Wang (2009). Alternating logistic regressions are Carey, Zeger and Diggle (1993). The failure of GEE under missingness that depends on observed responses, and the inverse-probability-weighted repair, are from Robins, Rotnitzky and Zhao (1995).

What is new here. Lemma 40.2.2 and its consequence — that a cluster-level random intercept cannot produce negative within-cluster correlation, so that the travel-mode data of Example (Four options per traveller) are outside the reach of any random-intercept GLMM — is elemen-

tary, and the association inequality behind it is standard, but we have not seen the two put together as a reason to prefer estimating equations. The efficiency comparison of Example (How much efficiency is at stake) separates within-cluster from cluster-level covariates, making the often-repeated advice “the working correlation hardly matters” precise: it hardly matters for cluster-level covariates and matters by a factor of nearly two for within-cluster ones.

Data and code. Grunfeld’s investment panel is the public-domain eleven-firm version distributed with `statsmodels`, also used in [Chapter 32](#) and [Chapter 33](#); its provenance is Grunfeld (1958), with the version history discussed by Kleiber and Zeileis (2010). The travel-mode choice data are the public-domain sample of 210 non-business trips between Sydney, Canberra and Melbourne from a 1987 intercity mode-choice study, also distributed with `statsmodels`; the sample over-represents the rarer modes, as Example (Four options per traveller) notes. The code in `code/ch40/` is written from the formulas in this chapter and checked against `statsmodels` where an equivalent fit exists; the simulations use fixed seeds.

40.6.3 References

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